

Capstone Program: PostgreSQL Mastery

12-Week Intensive Program — From SQL Basics to Production Expert

Program Overview

This 12-week capstone program transforms you from a SQL learner into a PostgreSQL expert capable of designing production systems, optimizing performance, and excelling in technical interviews at top-tier technology companies. The program combines structured daily study (1-2 hours/day), hands-on projects, mock interviews, and a final capstone project.

Total Commitment: 12 weeks, 1-2 hours/day, 5 days/week

Total Hours: ~120-160 hours

Outcome: Production-ready PostgreSQL proficiency + interview-ready

Prerequisites

Before starting this program, you should be comfortable with:

- Basic programming concepts (variables, loops, conditionals)
- General understanding of what a database is
- Command-line / terminal usage
- Ability to install software on your machine

Not required: Prior SQL experience (Week 1 starts from scratch)

Learning Outcomes

Upon completion of this program, you will be able to:

Technical Skills

- Write complex SQL queries including window functions, CTEs, and subqueries.
- Design normalized, production-quality database schemas.
- Implement PL/pgSQL stored functions and triggers.
- Configure and tune PostgreSQL for production workloads.
- Set up streaming replication and high-availability clusters.
- Diagnose and resolve performance bottlenecks using EXPLAIN ANALYZE.
- Implement security best practices (RLS, roles, encryption).
- Back up and restore PostgreSQL databases.
- Monitor and alert on key database metrics.

Career Skills

- Answer any PostgreSQL interview question with confidence.
 - Design systems under pressure (whiteboard/live coding).
 - Communicate database decisions to both technical and non-technical stakeholders.
 - Build a portfolio of 9 complete database projects.
 - Present a final capstone project to simulated reviewers.
-

Program Structure

PHASE 1: FOUNDATIONS (Weeks 1-2)

└ Week 1: Environment Setup + SQL Basics

└ Week 2: Intermediate SQL (JOINS, Aggregations)

PHASE 2: POSTGRESQL MASTERY (Weeks 3-6)

└ Week 3: Advanced SQL (Window Functions, CTEs)

└ Week 4: PostgreSQL Core (Types, Constraints, JSONB)

└ Week 5: Database Design (Normalization, ER Modeling)

└ Week 6: Indexes & Query Optimization

PHASE 3: INTERNALS & OPERATIONS (Weeks 7-9)

└ Week 7: Transactions, MVCC, Locking

└ Week 8: Advanced Features (Partitioning, FTS, Extensions)

└ Week 9: Production Ops (Monitoring, Security, Backup)

PHASE 4: ARCHITECTURE & CAREER (Weeks 10-12)

└ Week 10: Replication, HA, PgBouncer, Patroni

└ Week 11: Interview Preparation + Mock Interviews

└ Week 12: Capstone Project + Final Assessment

Weekly Time Commitment

Day	Activity	Time
Monday	Reading + Theory	60 min
Tuesday	Hands-on SQL Practice	90 min
Wednesday	Mini-Project / Exercises	90 min
Thursday	Mock Interview Questions	60 min
Friday	Review + Week Prep	45 min
Weekend	Optional: Extension challenges, blog posts	0-120 min

Tools Required

```
# Core
PostgreSQL 16 (local install)
psql (command-line client)
pg_dump / pg_restore

# Recommended GUI
DBeaver (free, cross-platform)
pgAdmin 4 (free, web-based)
TablePlus (paid, excellent UX)
```

```
# Optional but valuable
Docker (for multi-node setups)
git (for tracking your work)
VS Code with PostgreSQL extension
```

Evaluation Criteria

Your progress is evaluated across five dimensions:

1. SQL Query Quality (25%)

- Correctness: Does the query return the right results?
- Efficiency: Does it use appropriate indexes and avoid full scans?
- Readability: Is it formatted and commented clearly?
- Edge Cases: Does it handle NULLs, empty sets, and boundary values?

2. Schema Design (25%)

- Normalization: Is it in 3NF without unnecessary redundancy?
- Constraints: Are business rules enforced at the database level?
- Naming: Are conventions consistent and meaningful?
- Scalability: Would this design survive 100x data growth?

3. Performance Optimization (20%)

- Index selection: Right types for the right query patterns.
- EXPLAIN usage: Can you read and act on execution plans?
- Partitioning: Appropriate use of table partitioning.
- Configuration: Basic understanding of memory and parallelism settings.

4. Production Readiness (20%)

- Security: RLS, roles, least-privilege access.
- Monitoring: Key metrics identified and tracked.
- Backup strategy: Demonstrated backup and restore procedures.
- High Availability: Basic replication understanding.

5. Communication (10%)

- Interview responses: STAR format, concise, technically accurate.
- Design explanations: Can defend schema choices.
- Trade-off awareness: Understands when to denormalize, when not to.

Grading Scale

Score	Level	Description
90-100	Expert	Ready for senior DB engineer or DBA roles
75-89	Proficient	Ready for mid-level backend/data engineer roles
60-74	Competent	Ready for junior roles; continue practicing

Below 60	Developing	Repeat relevant weeks before proceeding
----------	------------	---

Capstone Project Options

In Week 12, choose ONE of the following:

1. **E-Commerce Platform:** Full-featured multi-tenant shop with payments, inventory, analytics, and RLS.
2. **Analytics Platform:** Time-series event store with funnel analysis, cohort retention, and real-time dashboards.
3. **Financial Ledger:** Double-entry bookkeeping system with auditing, interest calculation, and regulatory reports.
4. **Open Proposal:** Propose your own domain (must be approved by end of Week 10).

Week-by-Week Map to Repository Chapters

Week	Topic	Repository Chapters
1	Foundations	01_Fundamentals, 02_SQL_Basics
2	SQL Mastery	03_Intermediate_SQL
3	Advanced SQL	04_Advanced_SQL
4	PG Core	05_PostgreSQL_Core
5	DB Design	06_Database_Design
6	Indexes	07_Indexes, 08_Query_Optimization
7	Transactions	09_Transactions_Concurrency, 10_PostgreSQL_Internals
8	Advanced Features	11_Advanced_PostgreSQL
9	Production	12_Production_PostgreSQL, 14_Security, 15_Backup_Recovery
10	Replication & HA	13_Replication_HA
11	Interview Prep	19_Interview_Questions, 20_Interview_Tasks
12	Capstone	22_Projects

Community and Support

- Track your daily progress in a private journal or repo.
- Post your queries to the `#sql-review` channel for peer feedback.
- Use the `19_Interview_Questions` chapter for self-testing every Friday.
- If you get stuck, re-read the relevant chapter before asking for help.

Success Tips

1. **Type every query** — never copy-paste from the examples. Muscle memory matters.
2. **Break things on purpose** — drop indexes and observe query degradation.

3. **Read EXPLAIN output daily** — make it a habit, not an emergency tool.
4. **Build projects from memory** — after studying, close the notes and rebuild.
5. **Simulate pressure** — time yourself on interview questions. 5 minutes max per answer.
6. **Teach back** — explain concepts in your own words. If you can't, you don't know it.